

Scenario 4D: Short time-series stress test

MSSTNB fixed-gamma posterior performance under $T = 30$

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Table of contents

1 Purpose and main conclusion	2
2 Scenario design	2
2.1 What is changed in Scenario 4D?	2
2.2 Difference from S4A, S4B, and S4C	3
3 Data calibration	3
4 Fitting strategy	4
5 Fit-status classification	4
5.1 Replicate-level summaries	5
6 Posterior performance by subset	7
7 Regression recovery	7
8 Dispersion-parameter recovery	8
9 Latent-risk path recovery	9
10 Kappa recovery	9
11 Comparison with S3, S4A, S4B, and S4C	10
12 Why oracle diagnostics are not needed	11
13 What short time series does to the model	11
13.1 1. It reduces temporal replication but not count information	11
13.2 2. It preserves regression recovery	11
13.3 3. It preserves numerical stability	11
13.4 4. It weakens path-shape recovery	11
13.5 5. It leaves kappa recovery stable	11
14 Practical implications	12
15 Final conclusion	12

1 Purpose and main conclusion

Scenario 4D is designed as a **short time-series stress test** for the MSSTNB model. It keeps the Scenario 3 mean, exposure, covariate, spatial, dispersion, and dynamic latent-risk structure unchanged, but shortens the time horizon from $T = 100$ to $T = 30$.

The main conclusion is:

Shortening the time horizon to $T = 30$ reduces temporal information, but under this high-count and non-sparse design it does not create numerical instability. All 10 fixed-gamma fits are stable. Regression effects, the dispersion parameter, and the gamma-frailty component are recovered reasonably well. The main degradation is not instability, but weaker correlation-based latent-path recovery, especially for local temporal increments.

The key evidence is:

1. The data-generation check passes: average counts remain high, the zero proportion is zero, and the count CV remains close to the S3 reference scale.
2. The formal fixed-gamma fit gives 10 stable replicates and no soft-warning or numerical-instability replicate.
3. The posterior means of β_1 and β_2 are close to their truth values on average.
4. The posterior mean of r is slightly above the truth $r = 15$, but remains within a reasonable range.
5. Latent-path RMSE remains low, but $\text{cor}(\log \lambda)$ and $\text{cor}(\Delta \log \lambda)$ are weaker than in the longer $T = 100$ setting.

2 Scenario design

2.1 What is changed in Scenario 4D?

Scenario 4D changes only the number of time points:

Table 1: Official S4D short-time-series design.

quantity	value
Scenario ID	S4D_SHORT_TIME_T30
Stress source	Short time series
Reference T	$T = 100$
Stress T	$T = 30$
Ratio to reference	0.3
n1	9
Level-1 cells	270 per replicate
Exposure	Same as S3 reference
Mean structure	Same beta/phi/lambda structure as S3
Dispersion truth	$r = 15$
Gamma	Fixed at 0.8 in fitting
Replicates	10

Thus, the total number of level-1 area-time observations is reduced from $9 \times 100 = 900$ to $9 \times 30 = 270$. The goal is to isolate loss of temporal replication without also changing the count scale, exposure structure, or overdispersion level.

2.2 Difference from S4A, S4B, and S4C

The stress mechanism differs from the earlier Scenario 4 variants.

- S4A lowers the observation-level intensity and creates a sparse-count regime.
- S4B lowers and heterogenizes exposure, creating local information imbalance.
- S4C lowers r and creates strong overdispersion and heavier count tails.
- S4D shortens the temporal horizon, reducing temporal smoothing and repeated-time information.

Therefore, S4D asks whether the model can still separate regression effects, spatial effects, dispersion, frailty, and latent temporal risk when the time series is only 30% as long as the reference setting.

3 Data calibration

The official S4D $T = 30$ data passed all calibration checks. The count level remains high, the zero proportion is zero, and the count CV remains close to the reference scale.

Table 2: S4D data-calibration summary across 10 replicates.

metric	value
T	30
T/reference T	0.3
Level-1 cells per replicate	270
Mean count	204.927
Zero proportion	0.000
Count CV	1.012
Variance/mean	214.554
Total count	553,304
Maximum count	1,749
Max identified beta0	2.083
Max raw lambda	1.764
SD(log lambda)	0.493
SD(delta log lambda)	0.026
Data check	Passed

The data calibration shows that S4D does not duplicate S4A or S4C. The average count is about 205, the zero proportion is 0, and the count CV is about 1.01. Thus, the main reduction in information comes from fewer time points rather than from sparse observations or excessive overdispersion.

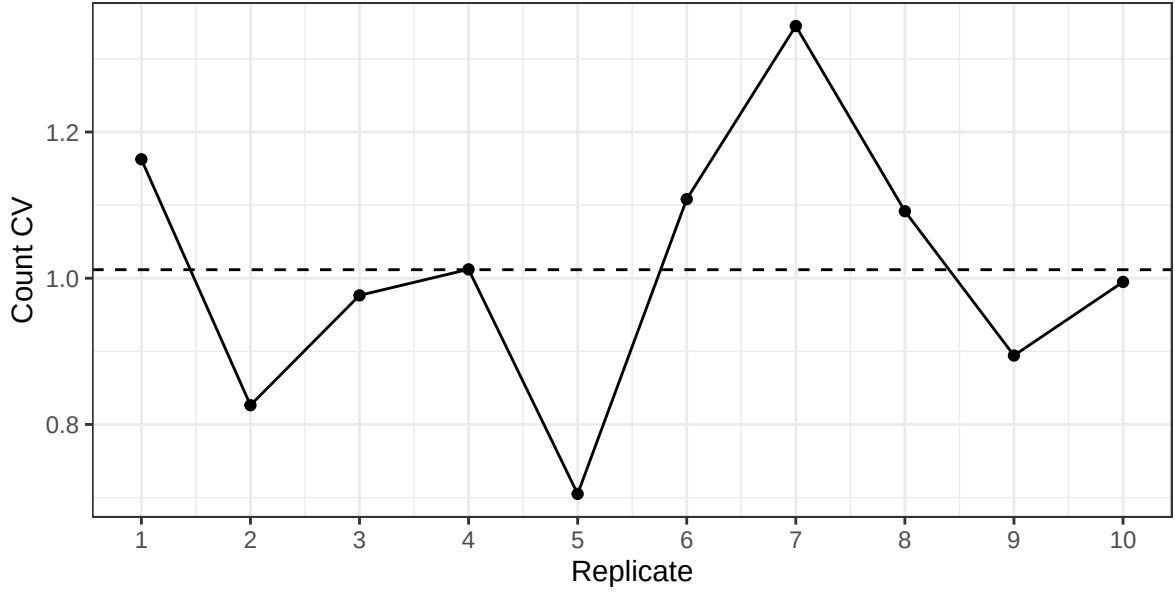


Figure 1: Count CV by replicate under the S4D short-time-series design.

4 Fitting strategy

S4D is fitted using the same fixed-gamma dynamic model structure used in the S4A–S4C stress diagnostics. The fitted model estimates

$$\beta_0, \beta_1, \beta_2, \phi, \tau_\phi, r, \kappa, \lambda.$$

The model fixes

$$\gamma = 0.8.$$

The model disables δ and ω updates. This fixed-gamma strategy isolates the short-time-series effect rather than re-testing weak identification of γ .

5 Fit-status classification

Each replicate is classified into one of three groups:

- **Stable:** no meaningful numerical guards and no extreme latent-process failure.
- **Soft warning:** small guard activity without severe parameter or latent-process failure.
- **Numerical instability:** large guard activity, extreme beta behavior, or extreme latent-path error.

Table 3: S4D fit-status counts.

status	n	prop
Stable	10	100%
Soft warning	0	0%
Numerical instability	0	0%

The result is 10 stable fits, 0 soft warnings, and 0 numerical-instability replicates. This is an important difference from S4A, S4B, and S4C: shortening T to 30 weakens temporal path-shape recovery but does not destabilize the sampler in this design.

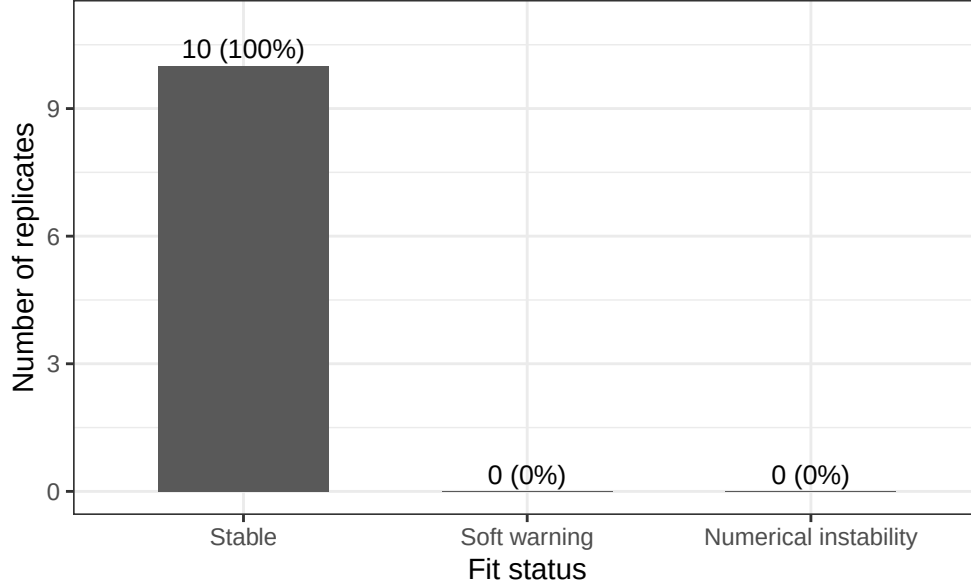


Figure 2: S4D fit-status classification across 10 replicates.

5.1 Replicate-level summaries

The replicate-level results are split into narrow tables to avoid over-wide PDF tables.

Table 4: Replicate-level S4D count summaries.

rep	status	mean	zero	cv	n_cells
1	Stable	180.2	0.000	1.163	270
2	Stable	208.1	0.000	0.826	270
3	Stable	177.9	0.000	0.977	270
4	Stable	223.0	0.000	1.012	270
5	Stable	243.2	0.000	0.705	270
6	Stable	224.0	0.000	1.108	270
7	Stable	225.6	0.000	1.345	270
8	Stable	188.5	0.000	1.092	270
9	Stable	220.8	0.000	0.894	270
10	Stable	157.9	0.000	0.995	270

Table 5: Replicate-level S4D regression and dispersion summaries.

rep	status	beta0	beta1	beta2	r
1	Stable	-1.683	0.479	-0.211	18.775
2	Stable	-1.489	0.473	-0.470	15.717
3	Stable	-1.667	0.493	-0.012	15.473
4	Stable	-1.351	0.539	-0.157	16.229
5	Stable	-1.441	0.509	-0.322	16.300
6	Stable	-1.607	0.471	-0.375	14.896
7	Stable	-1.516	0.474	-0.900	16.980
8	Stable	-1.498	0.523	-0.570	15.783
9	Stable	-1.450	0.454	-0.494	16.752
10	Stable	-2.073	0.527	-0.345	14.231

Table 6: Replicate-level S4D latent-risk path summaries.

rep	status	lam_rmse	loglam	cor_lam	cor_dlam
1	Stable	0.061	0.059	0.538	0.205
2	Stable	0.062	0.057	0.677	0.218
3	Stable	0.056	0.056	0.296	0.178
4	Stable	0.037	0.037	0.743	0.106
5	Stable	0.047	0.047	0.545	0.319
6	Stable	0.050	0.050	0.197	-0.136
7	Stable	0.048	0.048	0.660	0.202
8	Stable	0.040	0.040	0.574	0.128
9	Stable	0.039	0.039	0.151	0.032
10	Stable	0.068	0.068	0.407	0.080

Table 7: Replicate-level S4D kappa recovery summaries.

rep	status	truth_cv	post_cv	log_rmse	cor_log
1	Stable	0.269	0.233	0.134	0.868
2	Stable	0.262	0.237	0.122	0.887
3	Stable	0.261	0.243	0.110	0.908
4	Stable	0.249	0.228	0.110	0.906
5	Stable	0.262	0.241	0.095	0.930
6	Stable	0.272	0.250	0.110	0.918
7	Stable	0.252	0.220	0.119	0.883
8	Stable	0.249	0.226	0.109	0.910
9	Stable	0.259	0.234	0.115	0.894
10	Stable	0.278	0.242	0.142	0.872

Table 8: Replicate-level S4D numerical guard counts.

rep	status	beta_guard	kappa_guard	lambda_in	lambda_out
1	Stable	0	0	0	0
2	Stable	0	0	0	0
3	Stable	0	0	0	0
4	Stable	0	0	0	0
5	Stable	0	0	0	0
6	Stable	0	0	0	0
7	Stable	0	0	0	0
8	Stable	0	0	0	0
9	Stable	0	0	0	0
10	Stable	0	0	0	0

All guard counts are zero. Therefore, no oracle-lambda or oracle-lambda-plus-phi diagnostic is required for S4D.

6 Posterior performance by subset

Table 9: S4D posterior performance by fit-status subset: counts and regression summaries.

subset	n	T	cells	mean	zero	cv	beta1	beta2	r
All sampled	10	30	270	204.93	0.000	1.012	0.494	-0.386	16.114
Stable	10	30	270	204.93	0.000	1.012	0.494	-0.386	16.114
Stable+soft	10	30	270	204.93	0.000	1.012	0.494	-0.386	16.114
Unstable	0	NA	NA	NA	NA	NA	NA	NA	NA

Table 10: S4D posterior performance by fit-status subset: latent-risk and kappa summaries.

subset	n	loglam	cor_lam	cor_dlam	kappa_cv	logkap	cor_kap	lambda_guard
All sampled	10	0.050	0.479	0.133	0.235	0.116	0.898	0
Stable	10	0.050	0.479	0.133	0.235	0.116	0.898	0
Stable+soft	10	0.050	0.479	0.133	0.235	0.116	0.898	0
Unstable	0	NA	NA	NA	NA	NA	NA	NA

Since all replicates are stable, the all-sampled, stable-only, and stable-plus-soft-warning summaries are identical. The numerical-instability row is empty because no numerical-instability replicate occurred.

7 Regression recovery

Table 11: S4D beta recovery across stable replicates.

parameter	n	truth	post_mean	sd_reps	bias	rmse	coverage
beta0	10	-1.582	-1.578	0.203	0.004	0.009	1.00
beta1	10	0.500	0.494	0.029	-0.006	0.028	0.80
beta2	10	-0.400	-0.386	0.246	0.014	0.234	1.00

The regression results are strong. The posterior mean of β_1 averages about 0.494 against truth 0.5, and the posterior mean of β_2 averages about -0.386 against truth -0.4. Thus, unlike S4A/S4B and the failure replicate in S4C, the short- T design does not create substantial regression bias on average.

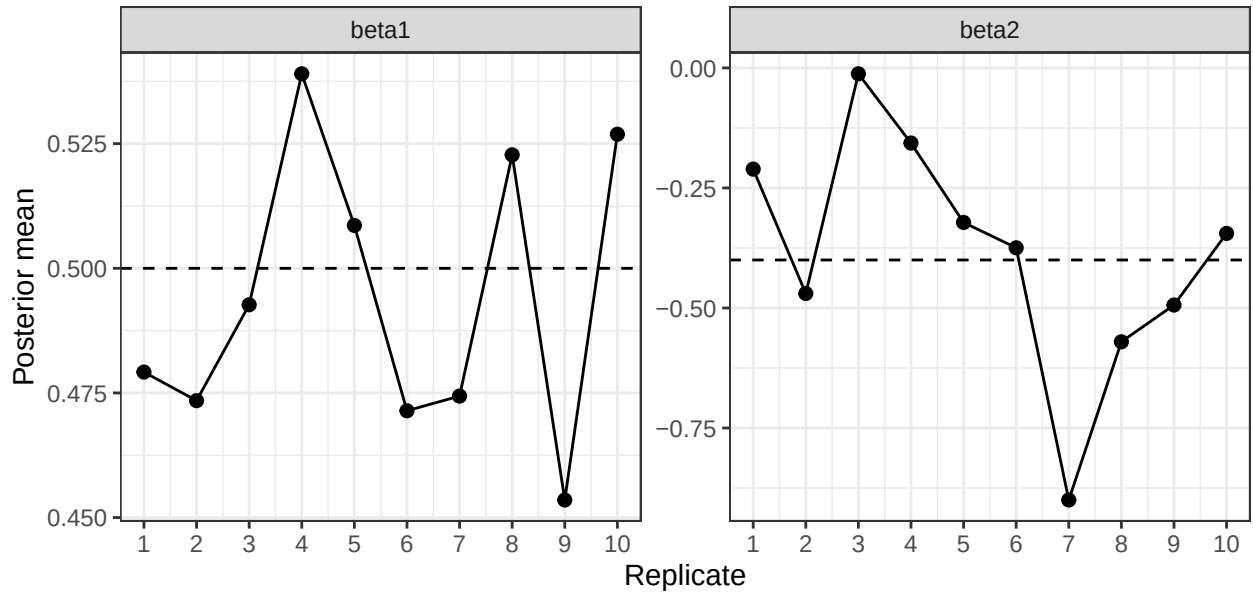


Figure 3: Posterior means for beta1 and beta2 by replicate. Dashed lines show the truth.

8 Dispersion-parameter recovery

Because S4D keeps the dispersion truth at $r = 15$, r recovery checks whether the shorter time series reduces information for the NB dispersion parameter.

Table 12: S4D dispersion-parameter recovery.

subset	n	r_truth	r_mean	r_sd	bias
Stable	10	15.000	16.114	1.247	1.114

The posterior mean of r is slightly above the truth but remains close enough to support stable dispersion recovery. This is reassuring because S4D has only 270 level-1 observations per replicate.

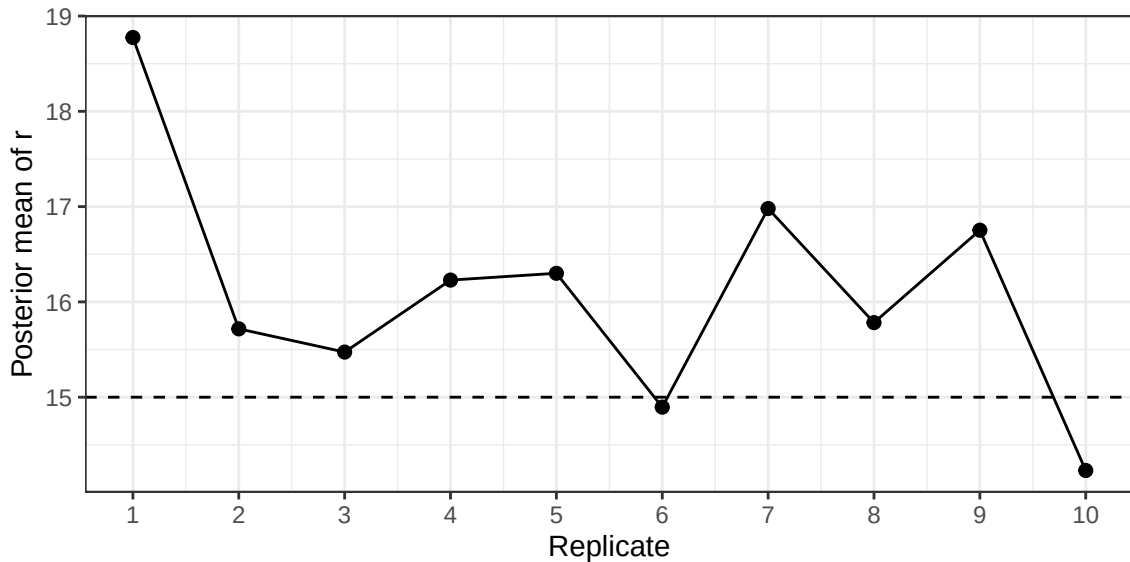


Figure 4: Posterior mean of r by replicate. The dashed line shows the truth $r = 15$.

9 Latent-risk path recovery

The main effect of S4D is on latent-risk path-shape recovery. The average log-lambda RMSE is small, but the average correlation-based metrics are weaker.

Table 13: S4D latent-risk path recovery across stable replicates.

metric	value
Lambda RMSE	0.051
Log-lambda RMSE	0.050
Cor(log lambda)	0.479
Cor(delta log lambda)	0.133

This pattern should be interpreted carefully. A low RMSE indicates that the overall magnitude of the latent-risk path is recovered reasonably well. The weaker correlations indicate that the shorter time horizon makes it harder to recover the precise temporal ordering, path shape, and local increments of $\log \lambda$.

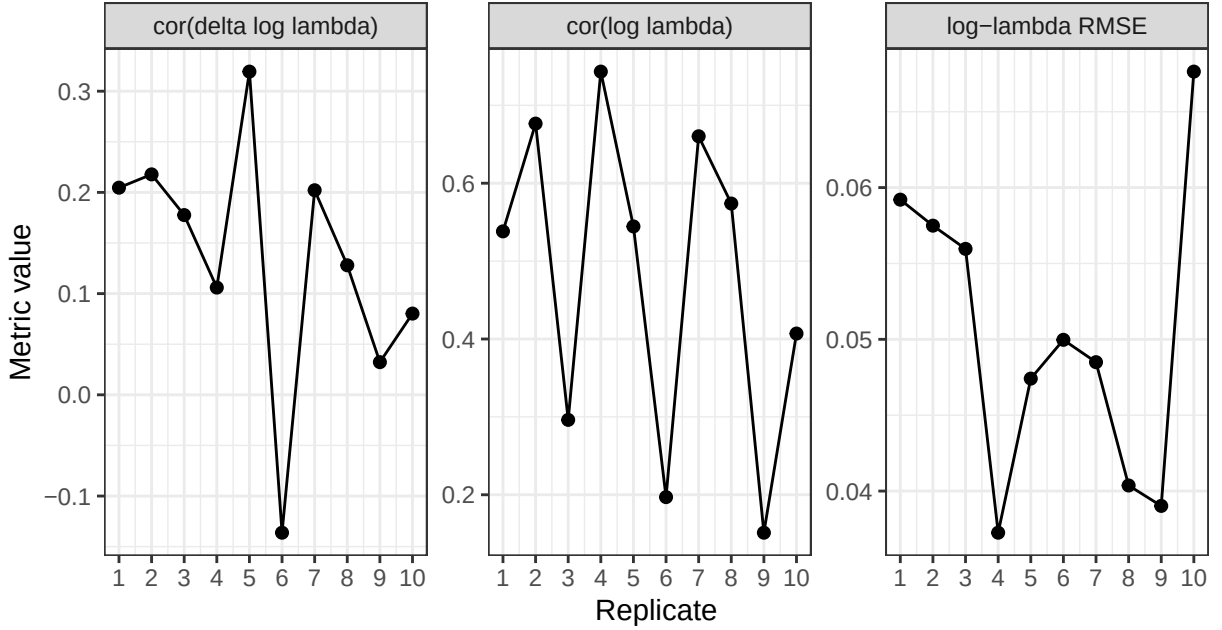


Figure 5: Latent-risk recovery metrics by replicate.

10 Kappa recovery

S4D also checks whether the gamma-frailty component remains stable when the number of time points is smaller.

Table 14: S4D kappa recovery across stable replicates.

metric	value
Truth kappa CV	0.261
Posterior-mean kappa CV	0.235
CV bias	-0.026
Log-kappa RMSE	0.116
Cor(log kappa)	0.898

Kappa recovery remains stable. The posterior-mean kappa CV is slightly lower than the truth CV, but

log-kappa RMSE is low and the average log-kappa correlation is high. This indicates that the short- T design does not meaningfully destabilize the kappa block.

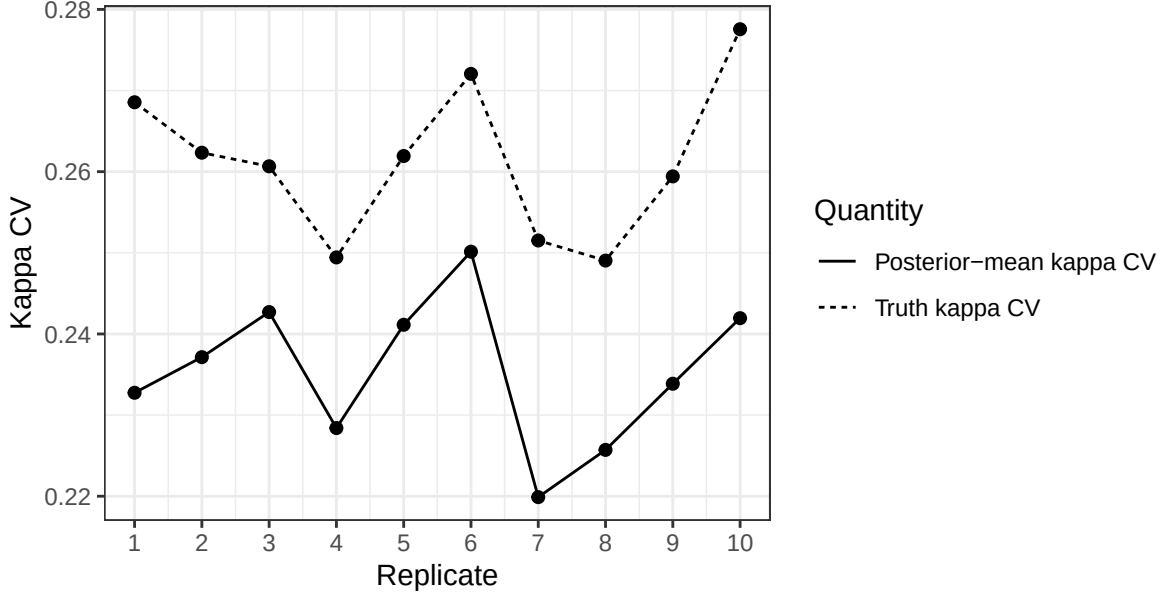


Figure 6: Kappa truth CV and posterior-mean kappa CV by replicate.

11 Comparison with S3, S4A, S4B, and S4C

Table 15: Scenario comparison: data scale and instability rate.

scenario	subset	n	mean	zero	instab
S3 control	All control	3	195.05	0.000	0%
S4A sparse	Stable+soft	8	3.17	0.223	20%
S4B exposure	Stable+soft	8	10.32	0.112	20%
S4C overdispersion	Stable	9	213.68	0.007	10%
S4D short time	Stable	10	204.93	0.000	0%

Table 16: Scenario comparison: beta2 and lambda recovery.

scenario	beta2	beta2_sd	loglam	cor_lam
S3 control	-0.408	0.009	0.047	0.734
S4A sparse	-0.675	2.528	0.206	0.435
S4B exposure	-0.479	1.800	0.185	0.363
S4C overdispersion	-0.220	0.513	0.075	0.721
S4D short time	-0.386	0.246	0.050	0.479

S4D is closer to the S3 control than to the low-information S4A and S4B stress regimes in terms of count scale and numerical stability. It also avoids the numerical failure seen in one S4C replicate. Its log-lambda RMSE is close to S3/S4C, but its average $\text{cor}(\log \lambda)$ is weaker than S3 and S4C, which is consistent with the intended short-time-series stress.

12 Why oracle diagnostics are not needed

For S4A, S4B, and S4C, oracle diagnostics were useful because at least one replicate showed numerical instability or severe latent-effect failure. In S4D, all 10 replicates are stable and all beta, kappa, and lambda guard counts are zero.

Therefore, oracle-lambda or oracle-lambda-plus-phi diagnostics are not required for S4D. The short-time-series effect is visible through weaker temporal correlation metrics rather than through sampler failure.

13 What short time series does to the model

13.1 1. It reduces temporal replication but not count information

The average count remains high and the zero proportion is zero. Therefore, S4D does not create sparse-count stress. The total amount of repeated temporal information is reduced because each area is observed at only 30 time points.

13.2 2. It preserves regression recovery

Both β_1 and β_2 are recovered well on average. This is an important contrast with S4A/S4B and the S4C failure replicate, where β_2 was more fragile.

13.3 3. It preserves numerical stability

All 10 replicates are stable. The beta, kappa, lambda-input, and lambda-output guard counts are zero across all replicates.

13.4 4. It weakens path-shape recovery

The main degradation is in $\text{cor}(\log \lambda)$ and especially $\text{cor}(\Delta \log \lambda)$. This indicates that short temporal horizons reduce information about the shape and local increments of the latent temporal path.

13.5 5. It leaves kappa recovery stable

Kappa recovery remains stable: posterior-mean kappa CV is slightly lower than the truth CV, but log-kappa RMSE is low and log-kappa correlation is high.

14 Practical implications

The S4D results suggest several practical implications for the MSSTNB model.

1. Shorter time series may be manageable when counts remain high and exposure is not sparse.
2. Temporal path-shape metrics should be interpreted cautiously when T is small.
3. RMSE and correlation measure different aspects of latent-risk recovery. A model can have low RMSE but weaker path-shape correlation.
4. Fit-status classification remains useful, but S4D shows that not every stress regime produces numerical failure.
5. Oracle diagnostics should be reserved for actual failures rather than applied automatically to every scenario.

15 Final conclusion

Scenario 4D confirms that reducing the time horizon from $T = 100$ to $T = 30$ creates a clean short-time-series stress regime without changing the count level, exposure structure, or overdispersion level. The resulting data remain high-count and non-sparse, with an average count of about 205 and zero observed zeros across the 10 replicates.

The fixed-gamma sampled-lambda fit is stable in all 10 replicates. Regression recovery is accurate on average, with β_1 close to 0.5 and β_2 close to -0.4. Dispersion recovery is also reasonable, with the posterior mean of r slightly above the truth 15. Kappa recovery remains stable and shows no numerical guard activity.

The main effect of the short time horizon is weaker latent-path shape recovery. The average log-lambda RMSE remains low, but $\text{cor}(\log \lambda)$ and especially $\text{cor}(\Delta \log \lambda)$ are weaker. Thus, S4D should be interpreted as a successful short-time-series stress test: the model remains numerically stable and recovers major parameters well, but short T limits the information available for detailed temporal path-shape recovery.

16 Recommended reporting language

The following paragraph can be adapted for the manuscript or internal report:

Scenario 4D evaluated the model under a short-time-series stress design by reducing the time horizon from $T = 100$ to $T = 30$ while preserving the reference mean, exposure, covariate, spatial, dispersion, and temporal latent-risk structure. This design reduced temporal replication without creating sparse counts or strong overdispersion. Across 10 fixed-gamma replicates, all fits were stable, with no beta, kappa, or lambda guard activity. Regression effects, the dispersion parameter, and kappa recovery remained accurate on average. The primary degradation appeared in correlation-based latent-risk path metrics: log-lambda RMSE remained low, but correlations for $\log \lambda$ and $\Delta \log \lambda$ were weaker. These results indicate that short time horizons mainly reduce information for detailed temporal path-shape recovery rather than causing numerical instability.